AI 500 - Practice Problem Set 3

Q1: KDE

Given the data points x = [1, 2, 3], use a Gaussian kernel with a bandwidth of h = 1 to estimate the probability density at x = 1.5.

Q2: KDE

We have three data points in 2D space: (1, 1),(2, 2),(3, 1). We're using a Gaussian kernel with a bandwidth of h = 1. Compute the probability density at the point (2, 1).

Q3: GMM

We have the data points x = [1, 2, 5, 6] and aim to fit a GMM with K = 2 components. The initial parameters are:

- $\pi_1 = 0.5$, $\mu_1 = 1$, $\sigma_1^2 = 1$
- $\pi_2 = 0.5$, $\mu_2 = 5$, $\sigma_2^2 = 1$

Compute the posterior probabilities $p(z|x = 2)$ and update μ_1 using one step of the Expectation Maximisation (EM) algorithm

Q4. Agglomerative Clustering

Given the following 5 data points in 2D space:

- A = (2, 3)
- B = (3, 3)
- C = (2, 2)
- D = (8, 7)
- E = (9, 7)

Using **single-linkage** (minimum distance between any two points in different clusters) as the distance measure, perform agglomerative clustering. Use Euclidean distance.

Tasks:

1. Calculate the initial distance matrix between all pairs of points
2. Show the first 3 merge steps of the agglomerative clustering algorithm

Q5. Divisive Clustering

Consider a dataset with 6 points: {P_1, P_2, P_3, P_4, P_5, P_6}

You are performing **divisive hierarchical clustering** (top-down approach). At each step, you split a cluster into two sub-clusters using some criterion.

Given the following hierarchy:

- Initial: C = {{P_1, P_2, P_3, P_4, P_5, P_6}}
- After split 1: C = {{P_1, P_2, P_3}, {P_4, P_5, P_6}}
- After split 2: C = {{P_1, P_2}, {P_3}, {P_4, P_5, P_6}}

Tasks:

1. Is this approach agglomerative or divisive? Explain.
2. What is the key characteristic of hierarchical clustering demonstrated here regarding cluster modifications?
3. If you wanted 3 clusters as your final result, at which step would you stop?

Q6. Agglomerative vs Divisive

Compare agglomerative and divisive hierarchical clustering approaches. Specifically address: a) The starting point of each approach b) The direction of hierarchy building c) Whether both require a fixed number of clusters k

Q7. Stopping Criteria

You are performing agglomerative clustering on a dataset and need to decide when to stop merging clusters.

Given the following stopping criteria definitions:

- Diameter of cluster C, dia(C): maximum inter-point distance in C
- Radius of cluster C with centroid c, rad(C): maximum distance of a point in C from centroid c
- Density of cluster C: mass (number of points) over volume

Tasks: a) Explain why hierarchical clustering might be stopped based on these quality measures rather than just quantity b) Give an example scenario where stopping based on diameter would be more appropriate than stopping based on quantity

Q8. (GMM)

a) GMM E-Step Computation

Consider a simple 1D dataset with 4 data points: x = [10, 12, 50, 52]

You are fitting a Gaussian Mixture Model with 2 Gaussians (K=2) to this data.

Current parameters (after some iterations):

- Gaussian 1: $w_1 = 0.5$, $\mu_1 = 11$, $\sigma_1 = 2$
- Gaussian 2: $w_2 = 0.5$, $\mu_2 = 51$, $\sigma_2 = 2$

Tasks:

1. For data point x = 10, compute $L_1(x)$ and $L_2(x)$
2. For data point x = 10, compute $P_1(x)$ and $P_2(x)$ (the responsibility/posterior probability)
3. Interpret what $P_1(10)$ and $P_2(10)$ tell us about this data point

Note: Use the formulas:

$$L_i(x) = w_i \cdot g(x \mid \mu_i, \sigma_i)$$

$$g(x \mid \mu_i, \sigma_i) = \frac{1}{\sigma_i \sqrt{2\pi}} \exp\left(-\frac{(x-\mu_i)^2}{2\sigma_i^2}\right)$$

$$P_i(x) = \frac{L_i(x)}{\sum_j L_j(x)}$$

b) GMM M-Step Parameter Update

Continuing from part (a), suppose after computing the E-step for all 4 data points x = [10, 12, 50, 52], you obtain the following posterior probabilities:

Data Point (x)	$P_1(x)$	$P_2(x)$
10	1.0	0.0
12	1.0	0.0
50	0.0	1.0
52	0.0	1.0

Tasks:

1. Update the weight w_1' for Gaussian 1
2. Update the mean μ_1' for Gaussian 1
3. Update the variance $\sigma_1'^2$ for Gaussian 1
4. What would w_2', μ_2', and $\sigma_2'^2$ be? (You may answer without full calculation)

Use the M-step formulas:

$$w_i' = \frac{\sum_x P_i(x)}{|X|}$$

$$\mu_i' = \frac{\sum_x x\, P_i(x)}{\sum_x P_i(x)}$$

$$\sigma_i'^2 = \frac{\sum_x P_i(x)\,(x - \mu_i')^2}{\sum_x P_i(x)}$$

c) EM Algorithm Understanding

Consider the EM algorithm for fitting Gaussian Mixture Models.

i) Explain why the problem of estimating GMM parameters is considered "hard" before using EM.

ii) What does the E-step assume and what does it compute?

iii) What does the M-step assume and what does it compute?

iv) The EM algorithm is iterative and may converge to local maxima. What does this mean for the initialization of parameters w_1, μ_1, σ_1, μ_2, σ_2?